

VASUDEV MALYAN

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[Personal Website](#) | [Google Scholar](#) | [ResearchGate](#) | [LinkedIn](#) | [ORCID](#)

EDUCATION

Indian Institute of Technology Bombay

Ph.D., Environmental Science and Engineering Department

Mumbai, India

07/2021 to 05/2026

Thesis: Calibration of Low-cost Particulate Matter Sensors Using Physics-based and Machine Learning Frameworks: From Laboratory Evaluation to Field Applications

Supervisor: Prof. Manoranjan Sahu

Indian Institute of Technology Bombay

M.Sc., Environmental Science and Engineering Department

Mumbai, India

07/2019 to 07/2021

Thesis: Techno-Economic Study of Air Pollution Control Technologies for Capturing Particulate Matter

Supervisor: Prof. Manoranjan Sahu

#Masters' seminar report, "Review of Air Pollution Control Technologies with a focus on Particulate Matter" published as book chapter in [Springer Nature](#)
9.68/10

Kirori Mal College, University of Delhi

B.Sc., Physical Sciences with Chemistry

Delhi, India

07/2016 to 05/2019

8.21/10

PROFESSIONAL EXPERIENCE

Ubiquedge Technologies Private Limited

Technical Expert — Air Quality Sensor Technologies

Hybrid, India

07/2026 to Present

Work: Leading the development of network-scale calibration models and source characterization approaches for ongoing low-cost PM sensor deployment at construction sites in Mumbai

Indian Institute of Technology Bombay

Research Associate

Mumbai, India

01/2026 to 05/2026

Work: Assessment of a physical-optical model for low-cost PM sensor calibration in the laboratory and field environments

Supervisor: Prof. Manoranjan Sahu

TEACHING EXPERIENCE

Teaching Assistant (TA) , Indian Institute of Technology Bombay	<i>Mumbai, India</i>
ES200 , Environmental Studies	01/2021 to 05/2021
ES209 , Data Analysis and Interpretation	2021, '22 and '24
ES672 , Air Pollution Control Technologies	2022, '23, '24
ES651 , Environmental Monitoring Laboratory	08/2023 to 11/2023
ES319 , Computational Laboratory for Environmental Engineers	01/2025 to 04/2025
Guest lecturer , Amicable Knowledge Solution University	<i>Madhya Pradesh, India</i>
Conducted 32 guest lectures for M.Sc. 1 st Year Environmental Science students on Environmental Chemistry, for the fundamentals of atmospheric chemistry	08/2024 – 11/2024
Guest lecture , Indian Institute of Technology Bombay	<i>Mumbai, India</i>
1-hour lecture on Air Pollution Control Technologies (ESP and Baghouse filters) in ES317 (Fundamentals of Air Pollution Science and Engineering)	2022

SKILLS

Programming	Python, R Studio, MATLAB, LaTeX
Tools	Machine Learning, Computational Fluid Dynamics
Software	QGIS, ArcGIS, Origin
Instrumentation	TSI Scanning Mobility Particle Sizer + Condensation Particle Counter (SMPS-CPC), Beta Attenuation Monitor (Met One BAM-1020), GRIMM Aerosol Spectrometer 11-D, GRIMM Optical Particle Sizer (OPS 1.109), TSI DustTrak Aerosol Monitor (DT II-8530), and Low-cost Particle Sensors (PurpleAir PA-II and Alphasense OPC-N2/OPC-N3), PALAS Aerosol Dust Generator, TSI Jet Nebulizer, Environmental Humidity and Temperature Control Chamber
Language	English, Hindi

AWARDS & FELLOWSHIPS

2023-25	Prime Minister Research Fellowship , IIT Bombay, India
2023	Late Shri Girish Vishnuprasad Desai Award for Academic Excellence , IIT Bombay, India
2023	Student Travel Award , Centre for Study of Science, Technology and Policy, India
2022	ASIC-India Travel Support , Air Quality Research Center, University of California, Davis, USA
2021-22	Ministry of Education Fellowship for Ph.D. , IIT Bombay, India
2019-21	Ministry of Education Fellowship for M.Sc. , IIT Bombay, India
2016	Fourth Place , Intel International Science and Engineering Fair, Phoenix, USA
2015	First Place , Initiative for Research and Innovation in STEM (IRIS) National Fair, India
2014	First Place , Initiative for Research and Innovation in STEM (IRIS) National Fair, India
2014	Special Award , Gujarat Council on Science and Technology, India
2013	First Place , Initiative for Research and Innovation in STEM (IRIS) National Fair, India

RESEARCH PROJECTS

IoT-Based Low-cost Sensor Network for Air Quality Monitoring in IIT Bombay 02/2022 to 04/2023

- Designed and implemented the IoT-based LCS network architecture across IIT Bombay
- Developed calibration algorithms for low-cost sensors, integrated them into the project software, and managed collaboration between the software development team and IIT Bombay stakeholders

Pilot study for assessment of reducing Particulate Air pollution in urban areas by using an air cleaning system (sometimes called as “Smog Tower”) 12/2021 to 04/2022

- Built an end-to-end analysis pipeline integrating reference monitors (E-BAM, Grimm OPS1.109, and TSI DustTrak) and LCSs to evaluate smog-tower impacts on hyperlocal air quality at two sites in NCT-Delhi
 - Designed automated worksheets to process and visualize filter differential pressure to track performance
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PUBLICATIONS

Citations: 152

h-Index: 7

i-Index: 5

Peer-reviewed Articles (First Author with #)

- 13#. **Malyan, V.** & Sahu, M. “Development and Assessment of a Physical-optical Calibration Model for Low-cost Sensors”. (*Manuscript under review*)
- 12#. **Malyan, V.**, Sahu, M., & Nawale, P. (2026). Design and Evaluation of a Calibration Chamber for Low-Cost PM Sensors—Part 2: Mass Concentration-Based Assessment. *ACS ES&T Air*, 3(6), 1677–1690. <https://doi.org/10.1021/acsestair.6c00161>
- 11#. **Malyan, V.**, Nawale, P., & Sahu, M. (2025). Design and Evaluation of a Calibration Chamber for Low-Cost PM Sensors—Part 1: Number Concentration-Based Assessment. *ACS ES&T Air*, 3(1), 129–141. <https://doi.org/10.1021/acsestair.5c00262>
10. Zaid, M., Nawale, P., Kumar, V., **Malyan, V.**, & Sahu, M. (2025). Investigating IoT-Based Low-cost Sensor Network for Real-Time Hyper-Local Air Quality Monitoring and Exposure Assessment. *Atmospheric Pollution Research*, 17(2), 102749. <https://doi.org/10.1016/j.apr.2025.102749>
- 9#. **Malyan, V.**, Kumar, V., Moni, M., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T. S., Fang, J., & Biswas, P. (2024). Assessing the Spatial Transferability of Calibration Models across a Low-cost Sensors Network. *Journal of Aerosol Science*, 181, 106437. <https://doi.org/10.1016/j.jaerosci.2024.106437>
8. Kumar, V., **Malyan, V.**, Sahu, M., & Biswal, B. (2024). Aerosol Sources Characterization and Apportionment from Low-Cost Particle Sensors in an Urban Environment. *Atmospheric Environment*. X, 22, 100271. <https://doi.org/10.1016/j.aeaoa.2024.100271>
- 7#. **Malyan, V.**, Kumar, V., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T. S., Fang, J., & Biswas, P. (2023). Calibrating low-cost sensors using MERRA-2 reconstructed PM2.5 mass concentration as a proxy. *Atmospheric Pollution Research*, 15(3), 102027. <https://doi.org/10.1016/j.apr.2023.102027>
6. Kumar, V., **Malyan, V.**, Sahu, M., Biswal, B., Pawar, M., & Dev, I. (2023). Spatiotemporal analysis of fine particulate matter for India (1980–2021) from MERRA-2 using ensemble machine learning. *Atmospheric Pollution Research*, 14(8), 101834. <https://doi.org/10.1016/j.apr.2023.101834>

5. Kumar, V., **Malyan, V.**, Sahu, M., & Biswal, B. (2023). Machine Learning Classification Model to Label Sources Derived from Factor Analysis Receptor Models for Source Apportionment. *Aerosol and Air Quality Research*, 23(7), 220386. <https://doi.org/10.4209/aaqr.220386>
4. Dharaiya, V. R., **Malyan, V.**, Kumar, V., Sahu, M., Venkatraman, C., Biswas, P., Yadav, K., Haswani, D., Raman, R. S., Bhat, R., Najjar, T. A., Jehangir, A., Patil, R. P., Pandithurai, G., Duhan, S. S., & Laura, J. S. (2023). Evaluating the Performance of Low-cost PM Sensors over Multiple COALESCE Network Sites. *Aerosol and Air Quality Research*, 23(5), 220390. <https://doi.org/10.4209/aaqr.220390>
3. Kumar, A., **Malyan, V.**, & Sahu, M. (2023). Air Pollution Control Technologies for Indoor Particulate Matter Pollution: A Review. *Aerosol Science and Engineering*, 7(2), 261–282. <https://doi.org/10.1007/s41810-023-00178-5>
- 2#. **Malyan, V.**, Kumar, V., & Sahu, M. (2023). Significance of sources and size distribution on calibration of low-cost particle sensors: Evidence from a field sampling campaign. *Journal of Aerosol Science*, 168, 106114. <https://doi.org/10.1016/j.jaerosci.2022.106114>
1. Kumar, V., **Malyan, V.**, & Sahu, M. (2022). Significance of Meteorological Feature Selection and Seasonal Variation on Performance and Calibration of a Low-Cost Particle Sensor. *Atmosphere*, 13(4), 587. <https://doi.org/10.3390/atmos13040587>

Book Chapter(s), and Policy Brief(s)

- 2#. **Malyan, V.**, Kumar, V., Sahu, M., Mayya, Y.S., Biswas, P., Venkatraman, C. (2023). Improving Air Quality Monitoring in G20 Countries through Low-Cost Sensor-Satellite Synergies. *T20 Policy Brief*.
1. Sahu, M., **Malyan, V.**, Mayya, Y.S. (2021). Technologies for Controlling Particulate Matter Emissions from Industries. In: Singh, S.P., Rathinam, K., Gupta, T., Agarwal, A.K. (eds) Pollution Control Technologies. Energy, Environment, and Sustainability. *Springer, Singapore*. https://doi.org/10.1007/978-981-16-0858-2_12

CONFERENCES (Platform with *, Poster with #)

- 10#. **Malyan, V.**, Nawale, P., & Sahu, M. December 2025. “Low-cost Particulate Matter Sensor Evaluation in a Calibration Chamber: Number and Mass Concentration Assessment” Mumbai, India. 14th Asian Aerosol Conference (AAC’25). [Lightning Talk + Poster Presentation]
- 9*. **Malyan, V.**, Kumar, V., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T.S., Fang, J., Biswas, P. February 2024. “Calibrating Low-cost Sensors using MERRA-2 Reconstructed PM_{2.5} Mass Concentration as a Proxy” Mumbai, India. National Environment Conference (NEC’24).
- 8*. **Malyan, V.**, Kumar, V., Moni, M., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T.S., Fang, J., Biswas, P. December 2023. “Optimizing Reference Co-location Sites for Improved Transferability of Calibration Models within a Low-cost Sensor Network in NCT-Delhi, India” Navi Mumbai, India. Indian Aerosol Science and Technology Association (IASTA) Conference.
- 7#. **Malyan, V.**, Kumar, V., Moni, M., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T.S., Fang, J., Biswas, P. October 2023. “Assessing the Spatial Transferability of Calibration Models across a Low-cost Sensors Network” Accra, Ghana. Air Sensor International Community Accra, Ghana Conference.

- 6#. **Malyan, V.**, Kumar, V., Sahu, M., Prakash, J., Choudhary, S., Raliya, R., Chadha, T.S., Fang, J., & Biswas, P. August 2023. "Distinct Seasonal Effects of Hygroscopicity on the Performance of Low-cost Particle Sensors" Bangalore, India. 5th India Clean Air Summit (ICAS).
 - 5#. **Malyan, V.**, Kumar, V., & Sahu, M. April 2023. "Significance of Sources and Size Distribution on Calibration of Low-cost Particle Sensors: Evidence from a Field Sampling Campaign" Vienna, Austria. The General Assembly 2023 of the European Geosciences Union (EGU).
 - 4*. **Malyan, V.**, Kumar, V., & Sahu, M. August 2022. "Fundamental Issues with Low-Cost Particle Sensors: Evidence from a Field Sampling Campaign" Bangalore, India. 2nd Air Sensors International Conference (ASIC). [Student Travel Award by UC Davis Air Quality Research Center]
 - 3*. Kumar, V., **Malyan, V.**, & Sahu, M. June 2022. "Significance of Meteorological Feature Selection and Seasonal Variation on Performance and Calibration of a Low-Cost Particle Sensor" Virtual. 12th Asian Aerosol Conference (AAC).
 - 2#. Kumar, A., **Malyan, V.**, & Sahu, M. June 2022. "Performance Comparison of different Particle Matter Control Technologies in Indoor Environment" Kuopio, Finland. 17th International Conference of the International Society of Indoor Air Quality & Climate (Indoor Air 2022).
 - 1#. Kumar, A., **Malyan, V.**, & Sahu, M. December 2021. "Comparative Analysis of different Indoor Particle Control Technologies" Virtual. 6th Indian International Conference on Air Quality Management (IICAQM).
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PERSONAL PROJECTS

Installation of Aerators on taps, Sustainability Cell, IIT Bombay 09/2019 to 04/2020

- One of five people working in the team
- Attending meetings, field surveys and communicating with authorities

A novel paper sensor as a diagnostic test for Multiple Sclerosis 08/2015 to 05/2016

- Designed an accurate and economically feasible paper-based sensor for the detection of multiple sclerosis
- Won fourth place in Intel International Science and Engineering Fair, Phoenix, USA
[Category: Translational Medical Science]

GGT Inhibition: Combating Multi-drug Resistance 08/2014 to 05/2016

- Developed a unique technique to inhibit a potential pathway of MDR in the bacterium
- Participated in International Science and Engineering Fair, Pittsburgh, USA [Category: Microbiology]
- Invited by the Indian Science Congress (ISC'14) to present the work
- Won a special award presented by the Gujarat Council on Science and Technology (GUJCOST'14)

Keratin Waste: An Effective Management 08/2013 to 05/2014

- Utilized keratin waste generated by the abattoir and the leather industry by microbial action of *B.licheniformis* ER15 to make a bio-fertilizer for better plant growth
 - Breakthrough to conventional waste management practices
 - Participated in Intel International Science and Engineering Fair, Los Angeles, USA [Category: Microbiology]
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ORGANIZATION OF CONFERENCES & PROFESSIONAL MEMBERSHIPS

2026	Lifetime Member , IIT Bombay Alumni Association, IIT Bombay, India
2023-24	Student Member , European Geosciences Union (EGU), Europe
2023-24	Student Member , Indian Aerosol Science and Technology Association (IASTA), India
2024-26	Organizer , Annual Symposium on Advanced Air Quality Sensing, ANTL, IIT Bombay, India
2024	Founder and Chief Organizer , Tatva, ESED Annual Fest, IIT Bombay, India
2024	Student Organizer , National Environment Conference (NEC'24), IIT Bombay, India

ACADEMIC SERVICE & POSITIONS OF RESPONSIBILITY

Department General Secretary, ESED Council <i>Environmental Science and Engineering, Indian Institute of Technology Bombay</i>	<i>Mumbai, India</i> 05/2023 to 04/2024
• Coordinating the student council and assisting students and professors in resolving academic issues • Conducting technical seminars, cultural events, industrial visits, and other formal/informal activities	
Internship Coordinator, Institute Placement Cell <i>Indian Institute of Technology Bombay</i>	<i>Mumbai, India</i> 09/2019 to 06/2020
• One of 35 members of the team catering to 1900+ students • Contacted companies and universities to hire interns from IIT Bombay during the placement period	
Alumni Secretary, Department Council <i>Environmental Science and Engineering, Indian Institute of Technology Bombay</i>	<i>Mumbai, India</i> 08/2021 to 05/2022
• Coordinated alumni engagement initiatives by organizing events and mentorship sessions	

REFERENCES

Professor Manoranjan Sahu

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Environmental Science and Engineering Department
Indian Institute of Technology Bombay, India
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Professor Srinidhi Balasubramanian

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Environmental Science and Engineering Department
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